

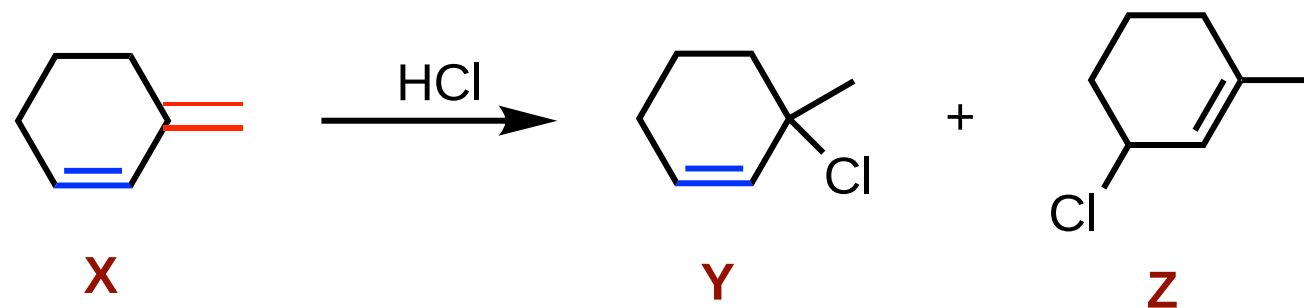
# CML-101

## Organic Chemistry

### **Tutorial-2**

Kinetic versus thermodynamic control: selected examples using MO theory (DAR, 1,3-butadiene addition, enolate alkylation, naphthalene sulfonation)

1) Addition of HCl to alkene **X** forms two alkyl halides **Y** and **Z**.

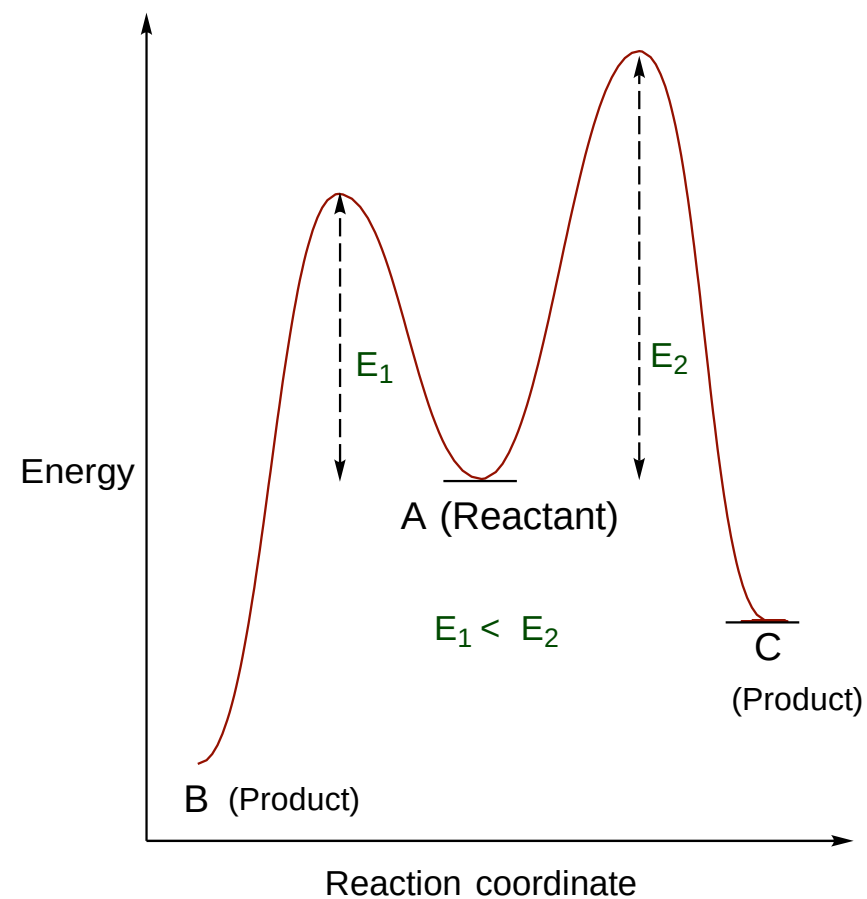
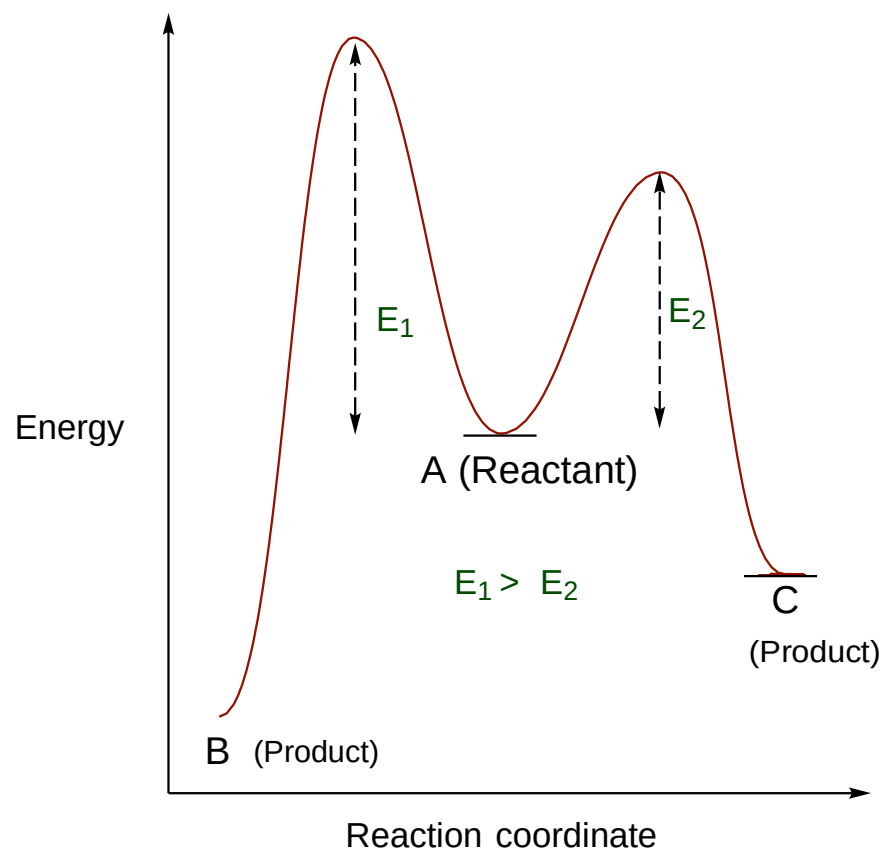


Q1: Label **Y** and **Z** as a 1,2-addition product or a 1,4- addition product

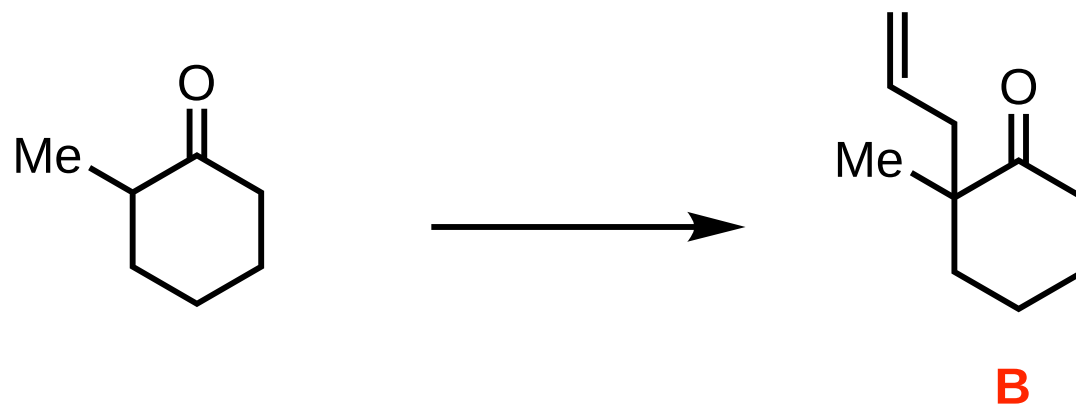
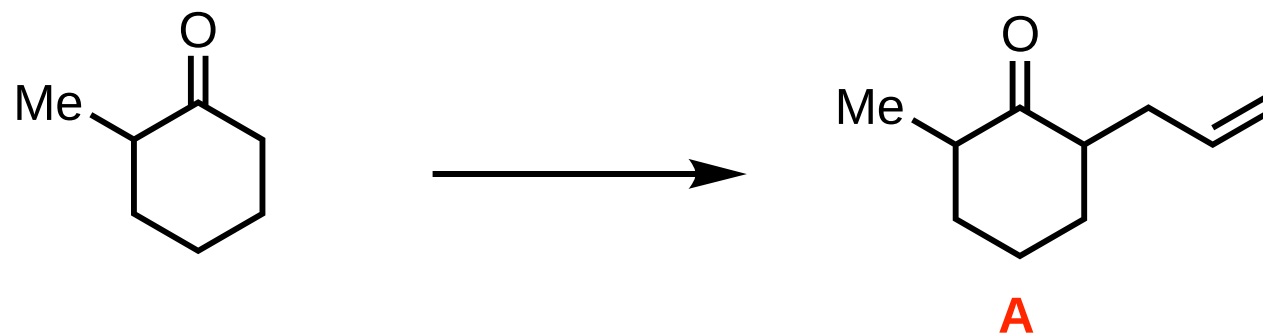
Q2: Label **Y** and **Z** as the kinetic or thermodynamic product and explain why.

Q3: Explain why addition of HCl occurs at the red **C=C**, rather than the blue **C=C**?

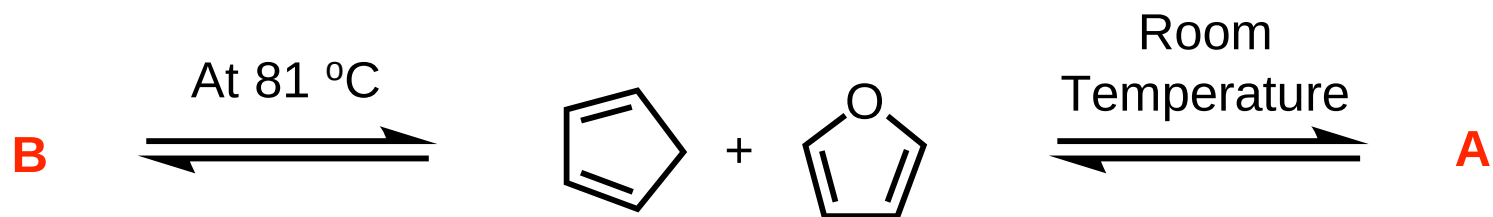
- 2) From the following potential energy diagrams
- Identify Kinetic and thermodynamic products
  - Explain the reason for kinetic and thermodynamic stability of the products.



3) Explain how to obtain A and B as major products in the following transformations?



4) Identify the products A and B in the following transformations?



- 5)
- (a) Distinguish between the transition state and an intermediate.
  - (b) Describe the importance of activation energy by drawing the reaction coordinate potential energy diagram for (i) an exothermic reaction, (ii) an endothermic reaction, and (iii) reaction of propene with HBr to yield 2-bromopropane and 1-bromopropane.
  - (c) Indicate the product of thermodynamic and kinetic control of reaction between (i) HBr and 1,3-butadiene (ii) Sulphuric acid and naphthalene and draw the potential energy curves for these reactions at the two specific temperatures.

6) How can 2-pentanone be converted into each of the following two compounds?

